7 In the 1930s, scientists discovered that when a slow neutron is absorbed by a uranium-235 nucleus, the nucleus splits into two smaller nuclei in a process known as nuclear fission. The products have a slightly smaller total mass than the original nucleus, and the missing mass is converted into energy according to Einstein's equation. Each fission reaction also releases, on average, 2.5 neutrons, which can induce further fission reactions. Under suitable conditions, this leads to a chain reaction. In a nuclear reactor, the chain reaction is carefully controlled to produce a continuous supply of energy. Fig. 7.1 illustrates a typical fission reaction.

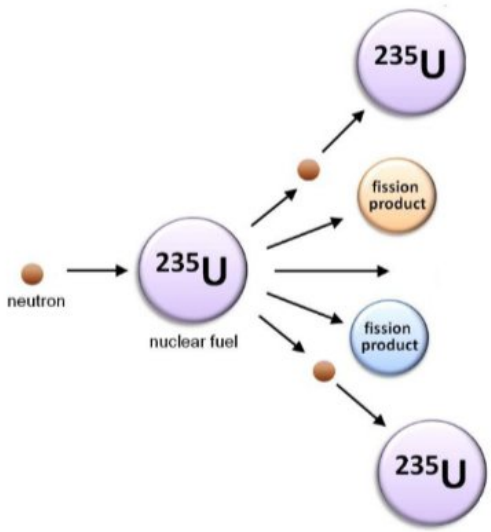


Fig. 7.1

When the conditions are suitable, a chain reaction can occur. If this chain reaction is not controlled, an explosion is likely to occur. However, if the chain reaction is controlled, as in a nuclear reactor, a source of continuous power may be created.

(a) (i) Explain what is meant by neutron-induced fission.

.....

.....

..... [1]

(ii) Explain what is meant by chain reaction.

.....

.....

..... [1]

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(iii) Suggest why, in an uncontrolled chain reaction where all neutrons are captured by Uranium-235 nuclei, the majority of the energy is released during the final stages of the fission of a sample of the uranium.

.....

.....

.....

..... [1]

(b) The induced fission reaction of Uranium-235 may be represented by a nuclear equation of the form

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{a}_{b}\mathrm{P} + {}^{c}_{d}\mathrm{Q} + (2 \text{ or } 3)\,{}^{1}_{0}\mathrm{n} + 198\mathrm{MeV}$$

The fission products P and Q have approximately equal masses. However, when any two nuclei are fissioned, the fission products may not be the same. If a large sample of Uranium- 235 is fissioned, many different fission products will be produced. The percentage amount of each fission product in the fissioned material is referred to as the percentage yield.

The variation with nucleon number of the percentage yield of different fission products is referred to as a 'fission yield curve' and is illustrated in Fig. 7.2.

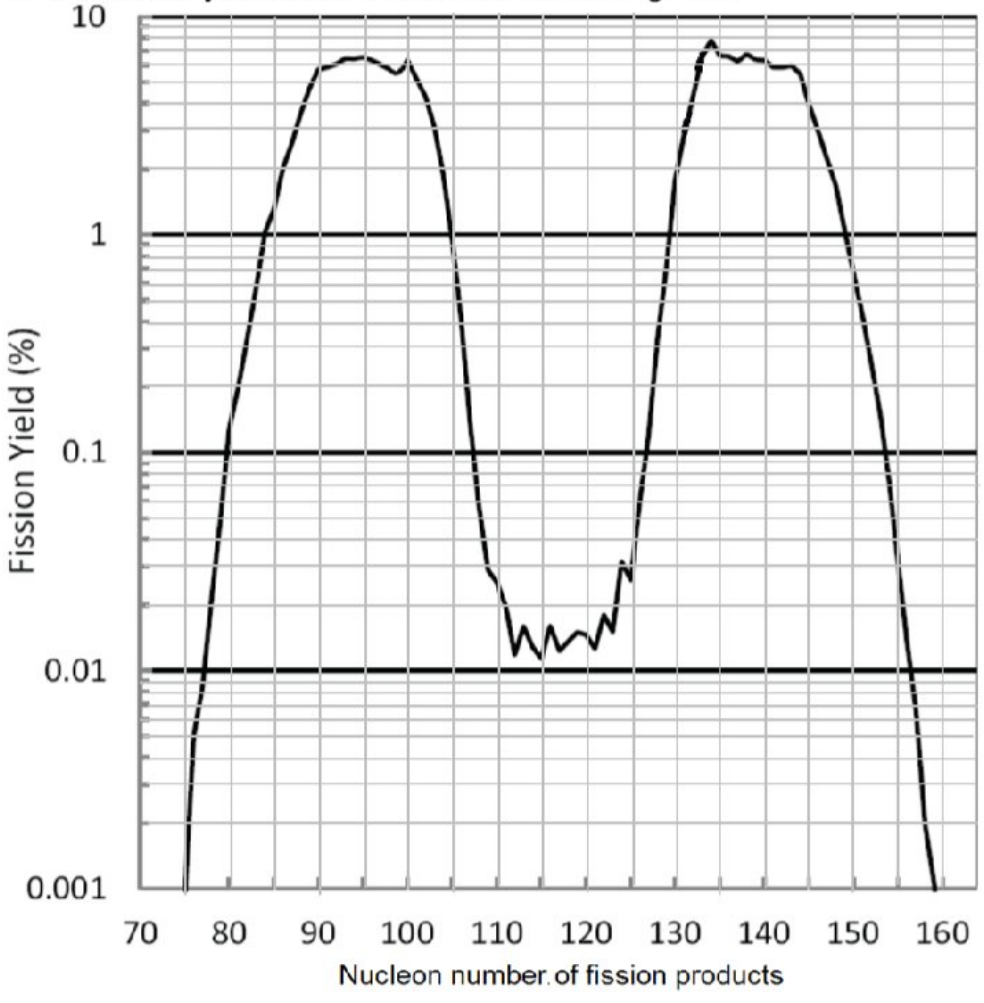


Fig. 7.2

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(i) State the proton number of the other fission product for the fission of one uranium if one of the products is $^{89}_{36}Kr$.

proton number = ... [1]

(ii) Suggest why the percentage yield is shown on a logarithmic scale.

.....

.....

..... [1]

(iii) Show that the percentage yield of Molybdenum-99 ($^{99}_{42}Mo$) is about 400 times more than those fission products having approximately equal masses.

[2]

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(c) The energy released during one fission reaction of a uranium nuclei occurs partly as kinetic energy of the fission products (167 MeV) and of the neutrons (5 MeV).

(i) Suggest one other mechanism by which energy is released in the fission reaction.

..... [1]

(ii) In a nuclear power station, 25% of the energy of the fission products is converted into electrical energy.
Calculate, for the fission of a mass of 1.0 kg of Uranium-235,

1. the number of nuclei in 1.0 kg of Uranium-235,

number of nuclei = ... [2]

2. the electrical energy generated,

energy = ... J [2]

3. the average power output, in megawatts, of the power station if the uranium is fissioned in a time of 24 hours.

power = ... MW [2]

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(d) The fission products are usually radioactive and give rise to a series of radioactive decay products. Each decay product has its own half-life. Two such fission products with their decay products and half-lives are shown below.

$$^{99}_{42}\text{Mo} \xrightarrow{67 \text{ hours}} {}^{99}_{43}\text{Tc} \xrightarrow{2 \times 10^5 \text{ years}} {}^{99}_{44}\text{Ru} \text{ (stable solid)}$$

$$^{140}_{54}\text{Xe} \xrightarrow{16 \text{ seconds}} {}^{140}_{55}\text{Cs} \xrightarrow{1.1 \text{ minutes}} {}^{140}_{56}\text{Ba} \xrightarrow{13 \text{ days}} {}^{140}_{57}\text{La} \xrightarrow{40 \text{ hours}} {}^{140}_{58}\text{Ce} \text{ (stable solid)}$$

Consider equal amounts of these two products.
Suggest why there are very different problems for the storage of this nuclear waste.

.....

.....

.....

..... [2]

[Total :16]